University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester I - UCSC AY22 [held in Sept./ Oct. 2025]

SCS 1306 — Linear Algebra

Two (2) Hours

Answer All Questions

Number of Pages = 16 Number of Questions = 4

237

To be completed by the candidate								
Index Number								

Important Instructions to candidates

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Write your index number CLEARLY on each and every page of this Question paper.
- This paper consists 3 structured questions and 15 MCQ based questions on **16** pages.
- Answer **ALL** questions. Questions 1 and 3 carry 30 marks each, while Questions 2 and 4 carry 20 marks each.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.

To be completed by the examiners	
1	
2	
3	
4	
Total	

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1. (a). Write down three types of square matrices and explain each with an example.

[3 marks]

(b). Consider the following linear system.

$$
\begin{aligned}
2x - y - 4z &= 0 \\
3x + 5y + 2z &= 5 \\
4x - 3y + 6z &= -16
\end{aligned}
$$

i. Write the augmented matrix of the above linear system.

[3 marks]

ii. Solve the above linear system using Gauss - Jordan Elimination Technique and find x, y and z.

[8 marks]

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(c). Explain Two (2) advantages of Reduced Row Echelon Form over Row Echelon Form in solving systems of linear equations.

[4 marks]

(d). Consider the following 3 x 3 matrix A.

$$A = \begin{pmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{pmatrix}$$

i. Find the elimination matrices E_{21}, E_{31}, and E_{32}.

[6 marks]

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ii. Using the elimination matrices E_{21}, E_{31}, and E_{32}, calculate the corresponding upper triangular matrix.

[6 marks]

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2. Consider the following matrix $A = \begin{pmatrix} 1 & 0 & 5 \\ -2 & 3 & 7 \\ 6 & -1 & 0 \end{pmatrix}$

(a). Calculate the determinant of A.

[4 marks]

(b). Find the cofactor matrix C of A.

[7 marks]

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(c). Determine the adjoint matrix of A.

[3 marks]

(d). Using the above results, compute the inverse of A.

[3 marks]

(e). By using a suitable method, justify whether the obtained inverse of A is correct.

[3 marks]

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3. (a). Define the Singular Value Decomposition (SVD) of a matrix.

[3 marks]

(b). Rectangular matrices are often converted into square matrices for analysis in linear algebra. Explain a standard method for converting a rectangular matrix into a square matrix. Illustrate the method with a suitable example.

[3 marks]

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(c). Consider the following matrix A.

$$A = \begin{pmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{pmatrix}$$

i. Find the eigenvalues and eigenvectors of A.

[5 marks]

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ii. Use the eigenvalues to compute the singular values of A.

[3 marks]

iii. Find the decomposed matrices U, Σ, and V^T.

[3 marks]

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iv. Verify that the original matrix can be reconstructed correctly from the product of the matrices U, Σ, and V^T.

[3 marks]

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4. Question 4 has 15 MCQ based questions. **(2 marks x 15 = 30 marks)**

- Each MCQ question will have 5 (five) choices with one correct answer.
- Mark the correct answers in the provided space at the last page of this paper.
- Cross or color the **best suite answer** among the given options.

(a). Why is Linear Algebra important for Computer Science students, and what are the advantages of learning it in the context of Computer Science?

i. It is only useful for solving simple arithmetic problems in programming.
ii. Its primary advantage is to replace calculus in all computer science applications.
iii. It is mainly required for understanding compiler design and operating systems.
iv. It provides the foundation for graphics, machine learning, data mining, and scientific computing.
v. Its sole purpose is to make mathematical proofs easier for computer science students.

(b). What does the rank of a matrix represent?

i. The total number of elements in the matrix.
ii. The number of linearly independent rows or columns.
iii. The largest entry (maximum value) in the matrix.
iv. The number of non-zero entries in the matrix.
v. The number of diagonal elements of the matrix.

(c). Which of the following statement about vector spaces is TRUE?

i. Every set of vectors in $\mathbb{R}^n$ automatically forms a vector space.
ii. The zero vector is optional in a vector space.
iii. A vector space must always have orthogonal basis vectors.
iv. The dimension of a vector space is always infinite.
v. A vector space must be closed under both vector addition and scalar multiplication.

(d). Which property of matrix multiplication makes it essential in computer science applications such as transformations and neural networks?

i. Matrix multiplication is commutative.
ii. Matrix multiplication always results in a diagonal matrix.
iii. Matrix multiplication reduces dimensionality automatically.
iv. Matrix multiplication guarantees eigenvalues are unchanged.
v. Matrix multiplication preserves linear transformations.

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(e). If A is a 2×3 matrix and B is a 3×4 matrix, what is the dimension of the product AB?

i. 2×2
ii. 3×3
iii. 2×4
iv. 4×3
v. Multiplication not possible

(f). Which of the following is a consequence of a matrix being singular?

i. It always has more columns than rows.
ii. Its eigenvalues must all be positive.
iii. Its rank equals the number of columns.
iv. Its determinant is zero, and it has no inverse.
v. It cannot represent a system of linear equations.

(g). Why is Singular Value Decomposition (SVD) important in applications?

i. It directly gives the inverse of any matrix.
ii. It decomposes a matrix into orthogonal components useful for dimensionality reduction.
iii. It is only used for solving differential equations.
iv. It eliminates the need for eigenvalue computation.
v. It only applies to diagonal matrices.

(h). Why is Linear Algebra fundamental for Machine Learning?

i. It only helps with storing large datasets.
ii. It eliminates the need for probability theory.
iii. It is only required for computer graphics.
iv. It guarantees faster training of all models without optimization.
v. It provides the framework for representing and manipulating data with vectors and matrices.

(i). What does it mean if a square matrix has determinant equals zero (0)?

i. The matrix has no eigenvalues.
ii. The matrix must be diagonal.
iii. The matrix contains only zeros.
iv. The rank is always full.
v. The matrix is singular.

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(j). What is the trace of a square matrix?

i. The product of diagonal entries.
ii. The number of independent columns.
iii. The sum of diagonal entries.
iv. The maximum eigenvalue.
v. The determinant.

(k). What does diagonalization of a matrix achieve?

i. Converts the matrix into a sparse representation.
ii. Ensures the matrix has full rank.
iii. Makes all off-diagonal entries zero by row reduction.
iv. Represents the matrix in terms of eigenvalues and eigenvectors.
v. Removes the need for matrix multiplication.

(l). Which of the following is TRUE about Cramer's Rule?

i. It can be applied to any rectangular matrix.
ii. It provides the exact solution using determinants if the coefficient matrix is invertible.
iii. It works only when the system has infinitely many solutions.
iv. It avoids determinant computation completely.
v. It applies only to diagonal matrices.

(m). Which of the following systems is not a linear system?

i. $\sqrt{5}\,x + 32\,y - 87\,z = 1$
ii. $100x - 0.001y + 7z = -15$
iii. $x_1 + 2x_2 + 3x_3 = 0$
iv. $x - 2y = 4 - 3z$
v. $xy + z - 7 = 4$

(n). Which statement is TRUE about the zero vector.

i. It can be a basis vector.
ii. It is linearly independent from every other vector.
iii. Adding it to any vector changes the vector.
iv. It has magnitude 1.
v. It has no magnitude and no direction.

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(o). In Gaussian elimination, what does a pivot represent?

i. Any zero in the matrix.
ii. The largest element in a row.
iii. The sum of diagonal entries.
iv. A non-zero leading entry used to eliminate other entries in its column.
v. The eigenvalue of the matrix.

MCQ Answer Sheet

(a)	(i)	(ii)	(iii)	(iv)	(v)
(b)	(i)	(ii)	(iii)	(iv)	(v)
(c)	(i)	(ii)	(iii)	(iv)	(v)
(d)	(i)	(ii)	(iii)	(iv)	(v)
(e)	(i)	(ii)	(iii)	(iv)	(v)
(f)	(i)	(ii)	(iii)	(iv)	(v)
(g)	(i)	(ii)	(iii)	(iv)	(v)
(h)	(i)	(ii)	(iii)	(iv)	(v)

(i)	(i)	(ii)	(iii)	(iv)	(v)
(j)	(i)	(ii)	(iii)	(iv)	(v)
(k)	(i)	(ii)	(iii)	(iv)	(v)
(l)	(i)	(ii)	(iii)	(iv)	(v)
(m)	(i)	(ii)	(iii)	(iv)	(v)
(n)	(i)	(ii)	(iii)	(iv)	(v)
(o)	(i)	(ii)	(iii)	(iv)	(v)

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